LECTURE 5

GROUP AND RING THEORY

We continue with establishing properties of sums of submodules.

Theorem 4.19. *If (N_i), for $1 \leq i \leq k$, is a family of R-submodules of a module M, then*

$$\sum_{i=1}^{k} N_i = \{x_1 + \cdots + x_k \mid x_i \in N_i\}.$$

Proof. Let $A = \{x_1 + \cdots + x_k \mid x_i \in N_i\}$. If $x_1 + \cdots + x_k$ and $y_1 + \cdots + y_k$ belong to A, then

$$(x_1 + \cdots + x_k) - (y_1 + \cdots + y_k) = (x_1 - y_1) + \cdots + (x_k - y_k)$$

also belong to A, because $x_i - y_i \in N_i$, for $1 \leq i \leq k$. Also, if $r \in R$, then

$$r(x_1 + \cdots + x_k) = rx_1 + \cdots + rx_k \in A,$$

because each rx_i is in N_i. Certainly $0 = 0 + \overset{k}{\cdots} + 0 \in A$, so $A \neq \varnothing$. Thus A is an R-submodule.

Further, if K is any R-submodule that contains each submodule N_i, then K contains all elements of the form $x_1 + \cdots + x_k$, where $x_i \in N_i$, since K is also an abelian group. Thus K contains A. Hence A is the smallest submodule containing each N_i. Therefore, from the definition of $\sum_{i=1}^{k} N_i$, we obtain

$$A = \sum_{i=1}^{k} N_i. \qquad \square$$

Remark 4.20. Let Λ be an infinite set. The sum $\sum_{i \in \Lambda} N_i$ of a family $(N_i)_{i \in \Lambda}$ of R-submodules of a module M is defined similarly as the submodule generated by $\bigcup_{i \in \Lambda} N_i$.

Following the proof of Theorem 4.19 (recall that for an abelian group, we only have closure when taking finite sums of elements), one obtains

$$\sum_{i \in \Lambda} N_i = \left\{ \sum_{\text{finite}} x_i \mid x_i \in N_i \right\},$$

where $\sum_{\text{finite}} x_i$ stands for any finite sum of elements of R-submodules N_i, for $i \in \Lambda$.

Definition 4.21. The sum $\sum_{i \in \Lambda} N_i$ of a family $(N_i)_{i \in \Lambda}$ of R-submodules of an R-module M is called a *direct sum* if each element x of $\sum_{i \in \Lambda} N_i$ can be *uniquely* written as $x = \sum_i x_i$, where $x_i \in N_i$, and $x_i = 0$ for all but finitely many i in Λ.

When the sum $\sum_{i \in \Lambda} N_i$ is direct, we write it as $\oplus_{i \in \Lambda} N_i$. Should it happen that Λ is a finite set $\{1, \ldots, k\}$, then the direct sum $\oplus_{i \in \Lambda} N_i$ is also written as $N_1 \oplus \cdots \oplus N_k$.

Each N_i in the direct sum $\oplus_{i \in \Lambda} N_i$ is called a *direct summand* of the direct sum.

We have already seen examples of direct sums of modules from Algebraic Structures, in the statement of the Fundamental Theorem of Finite Abelian Groups.

Theorem 4.22. *Let $(N_i)_{i \in \Lambda}$ be a family of R-submodules of an R-module M. Then the following are equivalent:*

(i) $\sum_{i\in\Lambda} N_i$ *is a direct sum;*
(ii) $0 = \sum_{i\in\Lambda} x_i \in \sum_{i\in\Lambda} N_i$ *implies* $x_i = 0$ *for all* i*;*
(iii) $N_i \cap \sum_{j\in\Lambda,\, j\neq i} N_j = (0)$*, for* $i \in \Lambda$*.*

Proof. (i) $\Rightarrow$ (ii): This follows from the definition of a direct sum.

(ii) $\Rightarrow$ (iii): Let $x \in N_i \cap \sum_{j\in\Lambda,\, j\neq i} N_j$. Then

$$x = \sum_{j\in\Lambda,\, j\neq i} n_j \in N_i,$$

thus

$$0 = \sum_{j\in\Lambda,\, j\neq i} n_j + (-x),$$

and by (ii) it follows that $x = 0$.

(iii) $\Rightarrow$ (i): Let $x = \sum_{i\in\Lambda} n_i$, with $n_i \in N_i$ and let also $x = \sum_{i\in\Lambda} m_i$, with $m_i \in N_i$. Observe also that both sums that express x are finite, as elements of the group M can only be a finite sum of other elements of the group. Then

$$\sum_{i\in\Lambda} n_i - \sum_{i\in\Lambda} m_i = 0$$

and so

$$n_i - m_i = \sum_{j\in\Lambda,\, j\neq i} n_j - m_j.$$

Hence

$$n_i - m_i \in N_i \cap \sum_{j\in\Lambda,\, j\neq i} N_j = (0),$$

and so $n_i = m_i$. Thus (i) follows. □

4.3. R-homomorphisms.

Definition 4.23. Let M, N be R-modules. A mapping $f : M \to N$ is an *R-homomorphism* of M into N if

(i) $f(x + y) = f(x) + f(y)$;
(ii) $f(rx) = rf(x)$,

for all $x, y \in M$ and $r \in R$.

We denote by $\mathrm{Hom}_R(M, N)$ the set of R-homomorphisms of M into N. If $M = N$, then f is called an *endomorphism* of M, and then the set $\mathrm{Hom}_R(M, M)$ is also denoted by $\mathrm{End}_R(M)$.

Let $f : M \to N$ be an R-homomorphism of an R-module M into an R-module N. Then f, in particular, is a group homomorphism. Thus

(a) $f(0) = 0$;
(b) $f(-x) = -f(x)$, for $x \in M$;
(c) $f(x - y) = f(x) - f(y)$, for $x, y \in M$.

Definition 4.24. Let $f : M \to N$ be an R-homomorphism of an R-module M into an R-module N. Then

(a) the set $\ker f = \{x \in M \mid f(x) = 0\}$ is called the *kernel* of f;
(b) the set $f(M) = \{f(x) \mid x \in M\}$ is called the *(homomorphic) image* of M under f and is also denoted by $\mathrm{im}\, f$.

It can be checked that $\ker f$ is an R-submodule of M, and that $\operatorname{im} f$ is an R-submodule of N (see Exercise Sheet 5). Further $\ker f = (0)$ if and only if f is one-to-one. If f is one-to-one, we say that M is isomorphic (or R-isomorphic) into N, or M is embeddable in N, or there is a copy of M in N, and we write it as $M \hookrightarrow N$.

If f is both one-to-one and onto, then f is an isomorphism and we say that M is isomorphic (or R-isomorphic) to N, and we write it as $M \cong N$. It can be shown that $\cong$ is an equivalence relation in the set of R-modules. (See Exercise Sheet 5.)

All linear transformations of vector spaces are examples of R-homomorphisms. Here are more examples.

Example 4.25. Let M be an R-module. Then the mapping $i : x \mapsto x$ of M onto M is clearly an R-homomorphism of M onto M. It is called the *identity endomorphism* of M. Similarly, the mapping $\overline{0} : x \mapsto 0$ of M into M is also an R-homomorphism of M into M. This is called the *zero endomorphism* of M. More generally one of course has the zero homomorphism between R-modules.

Example 4.26. Let M be an R-module over a commutative ring R, and let a be some fixed element of R. Then the mapping

$$f : x \mapsto ax, \quad \text{for } x \in M,$$

of M into M is an R-homomorphism of M into M. Indeed, for $x, y \in M$ and $r \in R$, we have

$$f(x + y) = a(x + y) = ax + ay = f(x) + f(y)$$

and

$$f(rx) = a(rx) = arx = rax = r(ax) = rf(x),$$

since R is commutative.

Note in particular that all linear transformations of vector spaces yield examples of R-homomorphisms.

Example 4.27. For $n \geq 2$ and $1 \leq i \leq n$, let $\pi_i : R^n \to R$ be the mapping defined by

$$\pi_i((x_1, \ldots, x_n)) = x_i \quad (\text{for } i \text{ fixed}).$$

Then π_i is an R-homomorphism of the R-module R^n onto the R-module R. This is called the *projection of R^n onto the ith component.*

Example 4.28. Let M be an R-module. Then $\operatorname{Hom}_R(M, M)$ is a subring of $\operatorname{Hom}(M, M)$.

Solution. Recall that $\operatorname{Hom}(M, M)$, the set of endomorphisms of M, regarded as an abelian group is a ring. Clearly $\operatorname{Hom}_R(M, M) \subseteq \operatorname{Hom}(M, M)$, and $\operatorname{Hom}_R(M, M)$ is non-empty since it contains the zero homomorphism.

Next, let $f, g \in \operatorname{Hom}_R(M, M)$. Since $\operatorname{Hom}(M, M)$ is a ring, we have that $f - g, fg \in \operatorname{Hom}(M, M)$ and so $f - g$ and fg satisfy part (i) of Definition 4.23. It remains to verify part (ii) of Definition 4.23. Let $x \in M$, and $r \in R$. Then

$$\begin{aligned}
(f - g)(rx) &= f(rx) - g(rx) \\
&= rf(x) - rg(x) \\
&= r(f(x) - g(x)) \\
&= r((f - g)(x)),
\end{aligned}$$

and

$$(fg)(rx) = f(g(rx)) = f(rg(x)) = r(f(g(x))) = r((fg)(x)).$$

Therefore, we have $f-g, fg \in \operatorname{Hom}_R(M, M)$. Hence $\operatorname{Hom}_R(M, M)$ is a subring of $\operatorname{Hom}(M, M)$.

Example 4.29. Let R be a ring with unity. Let $\mathrm{Hom}_R(R,R)$ denote the ring of endomorphisms of R regarded as a right R-module. Then $R \cong \mathrm{Hom}_R(R,R)$ as rings.

Solution. Consider the mapping $f : R \to \mathrm{Hom}_R(R,R)$, given by $f(a) = a^*$, where $a^*(x) = ax$ for $x \in R$. We first verify that $a^* \in \mathrm{Hom}_R(R,R)$. Let $x, y, r \in R$. Then

$$a^*(x+y) = a(x+y) = ax + ay = a^*(x) + a^*(y),$$

and

$$a^*(xr) = a(xr) = (ax)r = (a^*(x))r.$$

Thus a^* is an R-homomorphism of the right R-module R into itself. That is, $a^* \in \mathrm{Hom}_R(R,R)$. Note from the previous example that $\mathrm{Hom}_R(R,R)$ is a ring, and so for $a^*, b^* \in \mathrm{Hom}_R(R,R)$, we have $a^* + b^*, a^*b^* \in \mathrm{Hom}_R(R,R)$.

We now show that f is a ring homomorphism. Let $a, b \in R$. Then for any $x \in R$,

$$\begin{aligned}(a+b)^*(x) &= (a+b)(x)\\ &= ax + bx\\ &= a^*(x) + b^*(x)\\ &= (a^* + b^*)(x).\end{aligned}$$

Thus $(a+b)^* = a^* + b^*$. Similarly,

$$\begin{aligned}(ab)^*(x) &= (ab)(x)\\ &= a(bx)\\ &= a(b^*(x))\\ &= a^*(b^*(x))\\ &= (a^*b^*)(x),\end{aligned}$$

so $(ab)^* = a^*b^*$. Hence

$$f(a+b) = (a+b)^* = a^* + b^* = f(a) + f(b)$$

and

$$f(ab) = (ab)^* = a^*b^* = f(a)f(b).$$

Finally, we show that f is both one-to-one and onto. Let $a, b \in R$ and suppose $a^* = b^*$. Then $a^*(x) = b^*(x)$ for all $x \in R$. This gives $ax = bx$ for all $x \in R$. In particular, since $1 \in R$, we have $a = a \cdot 1 = b \cdot 1 = b$. Thus f is one-to-one. Now suppose $\phi \in \mathrm{Hom}_R(R,R)$. Let $\phi(1) = a$. We claim $\phi = a^*$, equivalently that a is a preimage of ϕ. Now for any $x \in R$,

$$\phi(x) = \phi(1x) = \phi(1)x = ax = a^*(x).$$

(remember: $\mathrm{Hom}_R(R,R)$ is regarded as a right R-module) hence $\phi = a^*$ as claimed. This proves that f is also an onto mapping.

We end this section by proving a result that has applications in subsequent sections.

Theorem 4.30. *Let $M = \oplus_{i=1}^k M_i$ be a direct sum of R-modules M_i. Then*

$$\mathrm{Hom}_R(M,M) \cong \begin{pmatrix} \mathrm{Hom}_R(M_1,M_1) & \mathrm{Hom}_R(M_2,M_1) & \cdots & \mathrm{Hom}_R(M_k,M_1)\\ \mathrm{Hom}_R(M_1,M_2) & \mathrm{Hom}_R(M_2,M_2) & \cdots & \mathrm{Hom}_R(M_k,M_2)\\ \vdots & \vdots & & \vdots\\ \mathrm{Hom}_R(M_1,M_k) & \mathrm{Hom}_R(M_2,M_k) & \cdots & \mathrm{Hom}_R(M_k,M_k)\end{pmatrix}$$

as rings.

The right-hand side is a ring T, say, of $k \times k$ matrices $f = (f_{ij})$ under the usual matrix addition and multiplication, where $f_{ij} \in \mathrm{Hom}_R(M_j, M_i)$.

Proof. Let $\phi \in \mathrm{Hom}_R(M, M)$. Let $\lambda_j : M_j \to M$ and $\pi_i : M \to M_i$ be the natural inclusion and projection mappings, respectively. Then $\pi_i \phi \lambda_j \in \mathrm{Hom}_R(M_j, M_i)$. Using the above notation, define a mapping $\sigma : \mathrm{Hom}_R(M, M) \to T$ by setting $\sigma(\phi)$ to be the $k \times k$ matrix $(\pi_i \phi \lambda_j)$ whose (i, j) entry is $\pi_i \phi \lambda_j$, where $\phi \in \mathrm{Hom}_R(M, M)$. Written differently, the (i, j) entry is $(\sigma(\phi))_{ij} = \pi_i \phi \lambda_j$.

We proceed to show that σ is an isomorphism. Let $\phi, \psi \in \mathrm{Hom}_R(M, M)$. Then

$$(\sigma(\phi + \psi))_{ij} = \pi_i(\phi + \psi)\lambda_j = (\pi_i \phi \lambda_j) + (\pi_i \psi \lambda_j) = (\sigma(\phi) + \sigma(\psi))_{ij},$$

and so

$$\sigma(\phi + \psi) = \sigma(\phi) + \sigma(\psi).$$

Further,

$$(\sigma(\phi)\sigma(\psi))_{ij} = \sum_{\ell=1}^{k} \big(\sigma(\phi)_{i\ell}\, \sigma(\psi)_{\ell j}\big)$$

by the definition of matrix multiplication. Hence

$$(\sigma(\phi)\sigma(\psi))_{ij} = \left(\sum_{\ell=1}^{k} \pi_i \phi \lambda_\ell \pi_\ell \psi \lambda_j\right) = \pi_i \phi \left(\sum_{\ell=1}^{k} \lambda_\ell \pi_\ell\right) \psi \lambda_j.$$

Since $\sum_{\ell=1}^{k} \lambda_\ell \pi_\ell = 1$, it follows that

$$(\sigma(\phi)\sigma(\psi))_{ij} = \pi_i \phi \psi \lambda_j = (\sigma(\phi\psi))_{ij},$$

and so

$$\sigma(\phi)\sigma(\psi) = \sigma(\phi\psi).$$

Therefore σ is a (ring) homomorphism.

To prove that σ is injective, let $\sigma(\phi) = (\pi_i \phi \lambda_j) = 0$. Then $\pi_i \phi \lambda_j = 0$, for $1 \leq i, j \leq k$. This implies that $\sum_{i=1}^{k} \pi_i \phi \lambda_j = 0$ for each $1 \leq j \leq k$. However as $\sum_{i=1}^{k} \pi_i = 1$, we obtain $\phi \lambda_j = 0$, for $1 \leq j \leq k$. Then from considering $0 = (\phi \lambda_j)\pi_j = \phi(\lambda_j \pi_j) = \phi$, we get $\phi = 0$, which proves that σ is injective. To prove that σ is surjective, let $f = (f_{ij}) \in T$, where $f_{ij} : M_j \to M_i$ are R-homomorphisms. Set $\phi = \sum_{i,j} \lambda_i f_{ij} \pi_j$. Then $\phi \in \mathrm{Hom}_R(M, M)$. By the definition of σ, we have $\sigma(\phi)$ is the $k \times k$ matrix whose (s, t) entry is $\pi_s(\sum_{i,j} \lambda_i f_{ij} \pi_j)\lambda_t = f_{st}$, because $\pi_p \lambda_q = \delta_{pq}$. Hence $\sigma(\phi) = (f_{st}) = f$. Thus σ is also surjective. $\square$

4.4. **Quotient modules.** Theorems analogous to the isomorphism theorems for groups and rings also hold for modules. Before we can obtain those theorems we need to define the concept of quotient modules (or factor modules).

Let N be an R-submodule of an R-module M. Recall that N is a (normal) subgroup of the abelian group M. So let $M/N = \{a + N \mid a \in M\}$ be the set of cosets of N in M. In M/N we define the binary operations as follows:

$$(a + N) + (b + N) = (a + b) + N, \qquad \text{for } a + N, b + N \in M/N,$$
$$r(a + N) = ra + N, \qquad \text{for } a + N \in M/N \text{ and } r \in R.$$

Following the construction of quotient groups, one can verify that these binary operations are well defined, and they turn the set M/N into an R-module. This module is called a *quotient module* (or a *factor module*) of M by N.

Theorem 4.31 (Correspondence Theorem)**.** *The submodules of the quotient module M/N are of the form U/N, where U is a submodule of M containing N.*

Proof. Let $f : M \to M/N$ be the canonical projection; i.e. $f(x) = x + N$, for $x \in M$. Let X be an R-submodule of M/N, and consider $U = \{x \in M \mid f(x) \in X\}$.

We claim that U is an R-submodule of M. Indeed, for $x \in N$, we have

$$f(x) = x + N = 0 + N \in X,$$

so $N \subseteq U$, and thus U is non-empty. Next, if $x, y \in U$ and $r \in R$, then as

$$f(x - y) = (x - y) + N = (x + N) - (y + N) = f(x) - f(y) \in X$$

and

$$f(rx) = rx + N = r(x + N) = rf(x) \in X,$$

we see that U is an R-submodule of M.

Finally observe that if $x \in X$, then there exists $y \in M$ such that $f(y) = x$, since f is an onto mapping. From the definition of U, we obtain $y \in U$. Hence $X \subseteq f(U)$. Clearly $f(U) \subseteq X$, and so $X = f(U)$. As $f(U) = U/N$, we obtain $X = U/N$, as desired. □

The proof of the following theorem on R-homomorphisms is analogous to that of the corresponding theorem for groups or rings.

Theorem 4.32 (First Isomorphism Theorem)**.** *Let f be an R-homomorphism of an R-module M into an R-module N. Then $M/\ker f \cong f(M)$.*

Proof. Consider the mapping $g : M/\ker f \to f(M)$ given by $g(x + \ker f) = f(x)$. We know from group theory that g is an isomorphism of abelian groups. So we only need to check that g respects the action of R. To this end, let $r \in R$. Then

$$g(r(x + \ker f)) = g(rx + \ker f) = f(rx) = rf(x) = rg(x + \ker f),$$

so g is an R-isomorphism of $M/\ker f$ onto $f(M)$. □

See Exercise Sheet 5 and 6 for the Second and Third Isomorphism Theorems. We end this section with some applications of the First Isomorphism Theorem.

Theorem 4.33. *Let A and B be R-submodules of R-modules M and N respectively. Then*

$$\frac{M \times N}{A \times B} \cong \frac{M}{A} \times \frac{N}{B}.$$

Proof. Define a mapping

$$f : M \times N \to \frac{M}{A} \times \frac{N}{B}$$

by $f((m, n)) = (m + A, n + B)$ for $m \in M$ and $n \in N$. It is a straightforward verification that f is an onto R-homomorphism. Now

$$\begin{aligned} \ker f &= \{(m, n) \mid (m + A, n + B) = (0 + A, 0 + B)\} \\ &= \{(m, n) \mid m \in A,\ n \in B\} \\ &= A \times B. \end{aligned}$$

Therefore, by the First Isomorphism Theorem,

$$\frac{M \times N}{A \times B} \cong \frac{M}{A} \times \frac{N}{B},$$

as required. □

Example 4.34. Let R be a ring with unity. An R-module M is cyclic if and only if $M \cong R/I$ for some left ideal I of R.

Solution. ($\Rightarrow$) Let $M = Rx$ be a cyclic module generated by x. Let $I := \mathrm{Ann}_R(x) = \{r \in R \mid rx = 0\}$, the *annihilator* of x in R. Then I is a left ideal of R. Define a mapping $f : R \to Rx$ by $f(r) = rx$ for $r \in R$. It is straightforward to check that f is an R-homomorphism that is also onto. Also,

$$\ker f = \{r \in R \mid rx = 0\} = I.$$

Hence, by the First Isomorphism Theorem, we have $R/I \cong Rx$.

($\Leftarrow$) For the converse, we note that the R-module R/I is generated by $1 + I \in R/I$; i.e. $R(1 + I) = R/I$. Hence R/I is cyclic.